

Riccardo Serraino

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Education

Alma Mater Studiorum - Università di Bologna, BSc in Automation Engineering — Sep 2022 —July 2025

Final Grade: 105/110
Thesis Grade: 4/4
GPA: 26.6/30.0
Thesis Title: Muscle Synergy-Based Analysis of Hand Motion Using sEMG and Optical Tracking

Tongji University (同济大学), BSc in Automation Engineering — Sep 2023 —July 2024

Exchange year in Shanghai (China) conducting the AlmaTong Double Degree programme.

Politecnico di Milano, MSc in Automation and Control Engineering — Sep 2025 —Present

Experience

Robotics Team Member - FRE2026, AIRLab Team, Politecnico di Milano — Mar 2026 —Jun 2026

Member of the AIRLab Team (AI and Robotics Lab) participating in FRE2026, contributing to the development of an autonomous agricultural robot for international field robotics competitions. Developed the perception subsystem described in the related project section.

Bachelor Thesis Student, University of Bologna — Apr 2025 —Jul 2025

Developed a dual-sensor Python framework integrating surface electromyography and optical motion tracking to analyze and classify human hand gestures.
Results contributed to robust, multimodal gesture recognition and laid foundations for applications in neuroprosthetics and adaptive human-machine interfaces.

Student Researcher (Bsc Level), University of Bologna — Feb 2025 —Apr 2025

Explored the control of robotic hands using sEMG signals by extracting muscle synergies through unsupervised learning methods such as NMF, PCA and Autoencoders.
Developed Python-based pipelines to implement and optimize synergy extraction algorithms, improving signal interpretation and control accuracy.

Technical Skills

Programming Languages: Python, C/C++, ROS2, SQL, MATLAB/Simulink, Verilog, SFC/ST, Docker

Robotics & AI: ROS2, YOLO, Computer Vision, Robot Perception, Machine Learning, Luxonis OAK-D, RViz, Foxglove Studio, Gazebo, Vicon, EMG armband

Certifications

IELTS certificate

Issuing Authority: British Council, Aug 2025

Chinese Language for Beginners

Issuing Authority: Politecnico di Milano, Dec 2025

Business Intelligence Data Analysis certificate

Completed a 50-hour Business Intelligence/Data Analyst program covering data cleaning, statistical analysis, dashboard creation, and transforming raw data into actionable insights.
Issuing Authority: Codecademy, Sep 2024

Projects

Insect Detection Stack for Agricultural Robotics (FRE2026) repository

Developed the perception pipeline for an autonomous agricultural robot competing in FRE2026.
Trained and deployed a YOLO11 model on a Luxonis OAK-D Pro camera for real-time detection of bees, butterflies and ladybugs in outdoor environments.
Integrated and validated the system using ROS2, Docker, RViz and Foxglove through field testing and competition trials, including automatic light signaling upon successful detections.
Tools Used: ROS2, Docker, YOLO11, Luxonis OAK-D Pro, RViz, Foxglove Studio, C++, Python

Muscle Synergy-Based Analysis of Hand Motion using sEMG and Optical Tracking repository

Work developed during the Bachelor Thesis at the Univeristy of Bologna.
Tools Used: Python, ROS1, Docker

Muscle Synergy Extraction and Analysis Pipeline repository

Work developed during the *Student Researcher* experience at the University of Bologna.
Tools Used: Python

Truss Bridge Dynamic Analysis repository

Finite element dynamic analysis of a steel truss bridge, including modal characterization and frequency response under seismic and moving loads. Identified resonance conditions and optimized the structure, achieving >20% vibration reduction with minimal mass increase using constraint-based design.
Tools Used: Verilog, GTKwave

Automatic Coffee Machine

Designed an automatic coffee machine with four beverage options and three coffee strengths. Implemented with sequential control using Sequential Function Chart (SFC) and Structured Text (ST). Integrated an HMI (for user interaction) and Generalized Actuator logic (for process management concerns).
Tools Used: CODESYS V3.5 SP20

RISC-V CPU core - Single Cycle repository

Designed a single cycle CPU based on RISC-V architecture, implementing most of the RV32I Base Integer instruction set.
Tools Used: Verilog, GTKwave

Character terminal based on RISC-V core and my personal custom IP repository

Designed an HDMI display module for RISC-V, mounted on the ICB bus of Hummingbird E203, that can display ASCII characters received from UART.
Tools Used: Verilog, GTKwave

Quadcopter Frame downloadable

Parametric CAD design of a quadcopter frame in Autodesk Fusion 360, optimized for lightweight structure, stiffness-to-weight ratio, and compatibility with standard components.
Tools Used: Fusion 360